STUDENT HANDOUT

Saylani Tech Blog

Basic knowledge, then the steps: install Sanity, learn the folders, connect it to Next.js, and publish a post.

Website name	Saylani Tech Blog
Sanity project	saylni-blog
Project ID	o2zj2owv
Dataset	production

1. Basic knowledge

This blog has two parts that work together.

Part	What it is	What you use it for
Next.js	The website	Pages a reader opens in the browser
Sanity	The content system	The form where you write and publish posts

This is called a **headless CMS**. The writing is not pasted inside a React file. You write it in Sanity. Next.js asks Sanity for the published posts and draws the page.

1. You write in Studio → 2. You press Publish → 3. Next.js asks Sanity → 4. The page shows the post

Words you need before the steps:

Word	Plain meaning
Project	Your Sanity space. Ours is named **saylni-blog**.
Project ID	The code uses this, not the name. Ours is `o2zj2owv`. It is public. It is not a password.
Dataset	The folder of content inside the project. Ours is `production`.
Schema	The form. This blog has one form, called `post`.

Studio	The editor, at `/studio` inside the same Next.js app.
GROQ	The question Next.js sends to Sanity to get posts.
Slug	The URL piece. “My new post” becomes `my-new-post`.
Draft	A saved post that readers cannot see.
Publish	The button that makes the post public.

> **Save is not Publish.** Save keeps a private draft. The website shows a post only after Publish.

2. What to install first

On your computer, install Node.js 20 or newer. It includes npm. Check it in a terminal:

```
node -v
npm -v
```

You also need a free account at sanity.io. This class already has a project:

- Name: **saylni-blog**
- Project ID: `o2zj2owv`
- Dataset: `production`

If you ever create your own project, the website at sanity.io/manage shows the project ID after you create it. Copy that ID. Do not type the project name into the code where the ID belongs.

3. Install Sanity in the Next.js project

Open a terminal in the project folder (the folder that already has `package.json`). Sanity is added to the Next.js app. It is not a second website.

Install the packages:

```
npm install sanity next-sanity styled-components @sanity/image-url @portabletext/react
```

Package	Why it is here
`sanity`	The Studio editor

`next-sanity`	Connects Studio and data fetching to Next.js
`styled-components`	Studio needs this to draw its screens
`@sanity/image-url`	Turns a Sanity image into a picture address
`@portabletext/react`	Turns the article body into paragraphs and headings

This project already has those packages in `package.json`. On a new computer you only run:

```
npm install
npm run dev
```

Then open `http://localhost:3000`. Leave the terminal open. Ctrl+C stops the server.

4. Folder structure

You do not need every file on day one. These are the ones that connect the blog.

```
headless/
  .env.local                      project ID and dataset
  sanity.config.js                Studio settings
  sanity.cli.js                   Sanity command-line settings
  src/
    app/
      (site)/
        page.js                   home page  →  /
        blog/page.js              post list  →  /blog
        blog/[slug]/page.js       one post   →  /blog/my-new-post
        about/page.js             about page →  /about
        layout.js                 header and footer
      studio/[[...tool]]/
        page.js                   editor route → /studio
        studio.js                 mounts Sanity Studio
    sanity/
      env.js                      reads the project ID
      structure.js                Studio sidebar (Posts)
      schemas/
        index.js                  list of forms
        post.js                   the post form
      lib/
        client.js                 the connection to Sanity
        queries.js                the GROQ questions
        posts.js                  functions the pages call
        image.js                  cover image addresses
    lib/site.js                   site name and menu
    components/blog/              the cards and the article body
```

Two folder rules:

- `(site)` does not appear in the URL. It only groups the public pages so they share a header and footer.
- `[slug]` means “any post address.” `[[...tool]]` lets Studio open its own inner pages under `/studio`.

5. Connect Sanity to Next.js

Connecting means four small jobs: tell Next.js the project ID, let the pages read posts, mount Studio, and describe the post form.

Step A — Put the project ID in `.env.local`

Create `.env.local` in the project root (copy `.env.example` if you need a start):

```
NEXT_PUBLIC_SANITY_PROJECT_ID=o2zj2owv
NEXT_PUBLIC_SANITY_DATASET=production
NEXT_PUBLIC_SANITY_API_VERSION=2026-10-17
NEXT_PUBLIC_SITE_URL=http://localhost:3000
```

Restart `npm run dev` after you change this file. Next.js reads it only at startup.

`src/sanity/env.js` reads those variables. The ID is safe to expose. Never put a Sanity write token in a `NEXT_PUBLIC_` variable. This blog only reads published posts, so it does not need a token.

Step B — The read connection

`src/sanity/lib/client.js` creates the Sanity client with the project ID, the dataset, and `perspective: "published"`. Pages never talk to Sanity by themselves. The path is:

page.js → getPosts() in posts.js → GROQ in queries.js → client.js → Sanity

The home page is an async Server Component. It can wait for the posts before it sends HTML:

```
const result = await getPosts();
```

On your computer the client does not cache, so a refresh shows a new post. A deployed site can take about a minute.

Step C — Mount Studio inside Next.js

`sanity.config.js` is the Studio setup. It uses the same project ID, the dataset, the post schema, and `basePath: "/studio"`. The file starts with `"use client"` because Studio runs in the browser.

`src/app/studio/[[...tool]]/studio.js` renders it:

```
import { NextStudio } from "next-sanity/studio";
import config from "../../../../sanity.config";

export default function Studio() {
  return <NextStudio config={config} />;
}
```

Open `http://localhost:3000/studio` and sign in.

Step D — Allow this computer (CORS)

The first time, Sanity may say the Studio is not connected. The project ID is already correct. Sanity is blocking the browser until you allow it.

1. Sign in at sanity.io/manage.
2. Open project **saylni-blog**.
3. Add CORS origin `http://localhost:3000`.
4. Turn on **Allow credentials**.
5. Save, then refresh `/studio`.

Do not put `/studio` on the end of the origin.

Step E — The post form

`src/sanity/schemas/post.js` is the form: title, slug, short summary, cover image, date, and body. `src/sanity/schemas/index.js` exports that one schema. `src/sanity/structure.js` puts **Posts** in the Studio sidebar.

After you change the schema file, refresh Studio. The public website still will not show a new field until the query asks for it and a component prints it.

6. Publish a post and see it

1. In Studio, open **Posts** and create a post.
2. Type a title.
3. Click **Generate** next to the slug.
4. Add a short summary, a cover image, and some body text.
5. Click **Publish**.
6. Refresh `http://localhost:3000`.

The post page is `http://localhost:3000/blog/` plus the slug. Example: `http://localhost:3000/blog/my-new-post`.

If nobody has published a post, the home page says “No posts yet.” That is correct. The site does not invent a sample article.

7. When something looks wrong

What you see	What to do

No posts yet, but you clicked Save	Click Publish, then refresh.
You published, and the page is still empty	Check the slug. Restart `npm run dev` if you just created `.env.local`.
Studio says it is not connected	Add CORS origin `http://localhost:3000` and allow credentials.
`node` is not recognized	Install Node.js, then open a new terminal.
A wrong slug	The not-found page. That is correct.

8. Practice

1. Point to the file that holds the project ID.
2. Point to the file that is the post form.
3. Point to the file that asks Sanity for the posts.
4. Publish one post with your name in the title and open it from the home page.

Answers: `.env.local` (read by `src/sanity/env.js`), `src/sanity/schemas/post.js`, and `src/sanity/lib/queries.js`.